

CO2 AT 1

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COURSE: JAVA PROGRAM
COURSE CODE: CSA0924

Q. A legacy inventory GUI uses a List component, a multi-line selectable list box, to let warehouse staff select multiple SKUs at once for a bulk price update, but the current code only handles single selection. Identify the AWT method/mode that enables multiple selection on a List component and how the event-handling code must change to retrieve all selected items instead of just one.

Ans:

Aim: To demonstrate how to enable **multiple selection** in a Java AWT List component and modify the event-handling code to retrieve and process **all selected SKUs** for a bulk price up

Algorithm:

- Start the Java AWT application.
- Create a List component containing SKU values.
- Enable multiple selection using setMultipleMode(true).
- Add the List component to the GUI.
- Create a button for the bulk price update.
- Add an event listener to the button.
- Check whether any SKU has been selected.
- Read the new price.
- Apply the new price to every selected SKU.
- Display the updated SKUs and the total number of selected items.
- Stop.

Pseudocode:

BEGIN

Create AWT List named skuList

Add SKU101

Add SKU102

Add SKU103

Add SKU104

Add SKU105

Enable multiple selection:

 skuList.setMultipleMode(true)

Create price input field

Create "Bulk Price Update" button

WHEN button is clicked:

 selectedSKUs = skuList.getSelectedItems()

 IF selectedSKUs is empty THEN

 Display "Please select at least one SKU"

 ELSE

 Read new price

 FOR each sku in selectedSKUs

 Update price of sku

 Display updated SKU

 END FOR

 Display total number of updated SKUs

 END IF

END

JAVA CODE:

```
import java.awt.*;
import java.awt.event.*;

public class InventoryMultiSelectGUI extends Frame implements ActionListener {
    List skuList;
    TextField priceField;
    TextArea output;
    Button updateButton;

    InventoryMultiSelectGUI() {
        setTitle("Inventory - Bulk SKU Price Update");
        setSize(700, 500);
        setLayout(new BorderLayout(10, 10));

        Panel topPanel = new Panel(new FlowLayout());
        Label titleLabel = new Label("Select SKUs for Bulk Price Update");
        topPanel.add(titleLabel);

        skuList = new List(10, true);
        skuList.add("SKU101 - Keyboard");
        skuList.add("SKU102 - Mouse");
        skuList.add("SKU103 - Monitor");
        skuList.add("SKU104 - Printer");
        skuList.add("SKU105 - Headset");
        skuList.add("SKU106 - Webcam");
        skuList.add("SKU107 - USB Cable");
        skuList.add("SKU108 - SSD");
        skuList.add("SKU109 - RAM");
        skuList.add("SKU110 - Speaker");
        skuList.setMultipleMode(true);

        Panel centerPanel = new Panel(new BorderLayout());
        centerPanel.add(skuList, BorderLayout.CENTER);

        Panel controlPanel = new Panel(new FlowLayout());
        Label priceLabel = new Label("New Price:");
        priceField = new TextField(10);
        updateButton = new Button("Bulk Price Update");
        updateButton.addActionListener(this);

        controlPanel.add(priceLabel);
        controlPanel.add(priceField);
        controlPanel.add(updateButton);

        output = new TextArea(10, 40);
        output.setEditable(false);
```

```

add(topPanel, BorderLayout.NORTH);
add(centerPanel, BorderLayout.CENTER);
add(controlPanel, BorderLayout.SOUTH);
add(output, BorderLayout.EAST);

addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        dispose();
    }
});

setVisible(true);
}

public void actionPerformed(ActionEvent e) {
    String[] selectedSKUs = skuList.getSelectedItems();

    if (selectedSKUs.length == 0) {
        output.setText("Please select at least one SKU.");
        return;
    }

    String newPrice = priceField.getText().trim();

    if (newPrice.isEmpty()) {
        output.setText("Please enter a new price.");
        return;
    }

    StringBuilder result = new StringBuilder();
    result.append("BULK PRICE UPDATE\n");
    result.append("-----\n");
    result.append("New Price: ₹").append(newPrice).append("\n\n");

    for (String sku : selectedSKUs) {
        result.append("Updated: ")
            .append(sku)
            .append(" -> ₹")
            .append(newPrice)
            .append("\n");
    }

    result.append("\n-----\n");
    result.append("Total SKUs Updated: ")
        .append(selectedSKUs.length);

    output.setText(result.toString());
}

```

```
}  
  
    public static void main(String[] args) {  
        new InventoryMultiSelectGUI();  
    }  
}
```

OUTPUT

BULK PRICE UPDATE

New Price: ₹1250

Updated: SKU101 - Keyboard -> ₹1250

Updated: SKU103 - Monitor -> ₹1250

Updated: SKU105 - Headset -> ₹1250

Total SKUs Updated: 3